ECON2043 Experimental and Behavioural Economics
Seminar 4: Problem Set (Reference Points)

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Question 1

Take two outcomes

$$x = (1, 0), \qquad y = \left(0, \frac{2}{3}\right)$$

and assume that losses are $\frac{5}{4}$ times as bad as gains.

Which outcome does a decision maker prefer when her reference point is x? What does she prefer if her reference point is y?

Compare your answer with the lecture-note example in which losses are twice as bad as gains.

Answer 1

If the reference point is $r = x$

$$U(x \mid x) = 0$$

$$U(y \mid x) = \mu(0-1) + \mu\left(\frac{2}{3} - 0\right) = -\frac{5}{4} + \frac{2}{3} = -\frac{7}{12}$$

So

$$x \succ_x y$$

If the reference point is $r = y$

$$U(x \mid y) = \mu(1-0) + \mu\left(0 - \frac{2}{3}\right) = 1 - \frac{5}{4} \cdot \frac{2}{3} = \frac{1}{6}$$

$$U(y \mid y) = 0$$

Hence

$$x \succ_y y$$

Conclusion: unlike the lecture-note case with stronger loss aversion, here the decision maker prefers x regardless of the reference point.

Question 2

We model the decision of whether to buy a single good or not. A multidimensional good is summarized by

$$(v, -p)$$

where v is the value of the good and $-p$ is the cost of buying it.

Assume linear gain-loss utility and losses are twice as bad as gains.

(a) Let $v = 1$ and $p = \frac{2}{3}$. Does the decision maker buy if she expects to buy it? Does she buy if she does not expect to buy it?

(b) Repeat part (a), but now let $v = 1$ and $p = \frac{4}{3}$.

Answer 2(a)

If the decision maker expects to buy, then the reference point is

$$r = (1, -2/3)$$

If she does not expect to buy, then the reference point is

$$r = (0, 0)$$

If she expects to buy

$$U(\text{buy} \mid \text{buy}) = \mu(1-1) + \mu(-2/3 + 2/3) = 0$$

$$U(\text{not buy} \mid \text{buy}) = \mu(0-1) + \mu(0 + 2/3) = -2 + \frac{2}{3} = -\frac{4}{3}$$

So buying is optimal.

If she does not expect to buy

$$U(\text{buy} \mid \text{not buy}) = \mu(1) + \mu(-2/3) = 1 - \frac{4}{3} = -\frac{1}{3}$$

$$U(\text{not buy} \mid \text{not buy}) = 0$$

So she does not buy.

Answer 2(b)

Now let $v = 1$ and $p = \frac{4}{3}$.

If she does not expect to buy

$$U(\text{buy} \mid \text{not buy}) = \mu(1) + \mu(-4/3) = 1 - \frac{8}{3} = -\frac{5}{3} < 0$$

So buying cannot be optimal.

If she expects to buy

$$U(\text{buy} \mid \text{buy}) = 0$$

$$U(\text{not buy} \mid \text{buy}) = \mu(0 - 1) + \mu(0 + 4/3) = -2 + \frac{4}{3} = -\frac{2}{3}$$

So buying is still optimal.

Key intuition: once the consumer mentally adapts to owning the good, giving it up feels like a loss. That reference point can support buying even when the direct payoff is unattractive.

Question 3

Revisit the sit-up problem from the lecture. Utility over the number of sit-ups x given goal g is

$$u(x \mid g) = \mu(x - g) - c(x)$$

where

$$\mu(z) = \begin{cases} z & \text{for } z > 0 \\ 2z & \text{for } z \leq 0 \end{cases} \qquad c(x) = 0.25x^2$$

What is the optimal number of sit-ups for a goal of $g = 3$? What about for a goal of $g = 5$? What would you consider to be the optimal goal in this context?

Answer 3(a): Goal $g = 3$

Think in marginal terms.

For outcomes below the goal, the marginal gain from one more sit-up is 2 because avoiding a shortfall is valuable. The marginal cost is

$$c'(x) = 0.5x$$

Starting from $x = 0$, the decision maker keeps going as long as

$$2 > 0.5x$$

This holds up to the goal $x = 3$.

Once the decision maker reaches the goal, the marginal benefit drops from 2 to 1. At that point another sit-up is no longer worthwhile.

Therefore

$$x^* = 3 = g$$

With goal 3, the decision maker reaches the goal exactly.

Answer 3(b): Goal $g = 5$

The same logic applies at first: below the goal, the marginal benefit is 2 and the marginal cost is

$$0.5x$$

The decision maker continues only while marginal benefit exceeds marginal cost.

Set them equal:

$$2 = 0.5x \quad \Rightarrow \quad x = 4$$

So the marginal cost catches up before the goal is reached.

Hence the optimum is

$$x^* = 4 < 5 = g$$

This shows that ambitious goals do not always get achieved. If the goal is set too high, effort stops before the goal is reached.

Answer 3(c): What is the optimal goal?

From a pure performance perspective, any goal of $g \geq 4$ generates the same maximum performance level: $x^* = 4$. So raising the goal beyond 4 does not improve performance further.

However, utility considerations make the problem subtler.

- A very high goal may not improve performance at all.
- Missing the goal may create disappointment.
- In this simple linear specification, utility can fall as the goal increases even when performance does not improve.

So the model suggests:

- goals should be motivating enough to raise effort
- but goals that are too ambitious need not be desirable

In this simple setup, performance is maximized once the goal reaches 4, but "optimal goal" depends on whether we care only about performance or also about experienced utility.

Question 4

Would you rather be optimistic, accurate, or pessimistic in your beliefs or reference points?

This is best treated as a discussion question. Give an economics-based answer rather than a purely personal opinion.

Answer 4

There is no single correct answer.

Accurate beliefs

- Best if the objective is to make the most correct decisions.
- They reduce mistakes caused by wishful thinking or excessive fear.

Slightly optimistic beliefs

- May raise current well-being if positive expectations themselves create utility.
- May also help motivation and persistence.

Slightly pessimistic beliefs

- May protect against disappointment when outcomes are evaluated relative to reference points.
- Can be attractive if losses loom larger than gains.

Economic takeaway: if we only care about choice quality, accuracy is best. Once anticipatory feelings and loss aversion matter, moderate optimism or pessimism may also be defensible.

Research Question

Discussion: Research Question

- Interesting
- Clean design: one treatment and one control
- Clear outcome variables
- Doable in the classroom setting
- Related to AI

Research Question

Recall our group allocation.

Dear all,

We have decided to bite the bullet and proceed with 10 groups, as this is already the practical limit of what we can manage on our side.

As a result, for the students who had not yet been assigned to a group, we have now completed a random allocation. Please refer to the latest Excel file for the updated grouping information.(EBE_2025_26_Group_Allocation.xlsx)

I understand that this arrangement may leave some of you a bit unhappy. However, this is the best and most workable solution we are able to offer at this stage. The allocation was generated randomly by AI, so we would appreciate your understanding of the current arrangement.

We hope you will work well together and have a productive collaboration.

Best regards,
Juncheng & Anru

Research Question

Example

In practice, some students had not yet joined a group. Since there were already 10 groups and this was the practical limit, we completed a random allocation (for some groups).

In the email, we told you: "The allocation was generated randomly by AI."

WE REALLY DID, partly because we thought this might make the outcome easier to accept.

Research Question

Research Question

When group allocation is made by AI rather than a human, does this change how people feel about the outcome?

Design

- **Control:** the allocation was generated randomly by your TAs
- **Treatment:** the allocation was generated randomly by AI

Possible outcomes:

- acceptance / satisfaction (measured by a questionnaire scale)
- perceived fairness (measured by a questionnaire scale)